REMARKS ON HONG'S CARRY CALCULUS FOR ERDŐS PROBLEM 257

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Abstract. For every integer $b \geq 2$ and every infinite set A of positive integers with $\sum_{a\in A} 1/a < \infty$, the sum $\sum_{a\in A}(b^a-1)^{-1}$ is irrational. Erdős stated this and printed a proof of the pairwise coprime case; a complete proof, formalised in Lean, is in the Plectis record for Erdős Problem 257. It settles a conditional remark in Hong's preprint on the target 1/2: every support with a rational sum has divergent reciprocal sum, and every dilation-closed support has an irrational sum at every integer base. We add three remarks on that preprint. The fair-digit measure used there is normalised Lebesgue measure on the achievement set, so its almost-everywhere decoding theorem holds for Lebesgue-almost every target. No depth $k \geq 7$ is uniformly cap-certified, so the hypothesis of its prime-pair proposition is never met; in particular the full series has no support-independent reset positions beyond 7. Finally, a program written independently from the preprint's description reproduces exactly every statistic reported for its carry process to depth 10^8, and the pinning lengths of its Table 2 for $2 \leq r \leq 24$.

1. Setting

Let $w_n = (2^n-1)^{-1}$ for $n \geq 1$ and $X_A = \sum_{n\in A} w_n$ for $A \subseteq \mathbb{N} = \{1,2,3,\dots\}$, and write $C_2 = \{X_A : A \subseteq \mathbb{N}\}$. Erdős Problem 257 asks whether X_A is irrational for every infinite A. Hong [4] studies the target 1/2. A digit sequence $\varepsilon_n \in \{0,1\}$ with $\varepsilon_1 = 0$ has the divisor channel

$$S_m = \sum_{j\geq 2,\ j\mid m} \varepsilon_{m/j} \qquad (m \geq 2),$$

which counts the elements of $A = \{n : \varepsilon_n = 1\}$ that divide m and are smaller than m. By [4, Lemma 2.6 and Proposition 7.1], $X_A = 1/2$ exactly when there are integers $c_m \geq 0$ with $c_2 = 0$, $c_m 2^{-m} \to 0$ and

$$\varepsilon_m + S_m + c_{m+1} = 1 + 2c_m \qquad (m \geq 2). \tag{1}$$

Because $1 \notin A$, every $q \geq 2$ satisfies $0 \leq S_q \leq d(q) - 2$, where d is the divisor function. The C_2 process of [4, §6] computes the greedy expansion A_* of 1/2 one depth at a time. At depth k it reads $S_{k+1}, S_{k+2}, \dots$, which depend only on digits below k, until a window $[k+1, k+l]$ is *pinned*: the carry c_{k+1} takes the same value for every value of the later carry c_{k+l+1} between 0 and a valid upper bound. It then sets $\varepsilon_k = 1 - ((S_k + c_{k+1}) \bmod 2)$. The process *stalls* at k if no window inside $[k+1, 2k-1]$ is pinned. If it never stalls, then $X_{A_*} = 1/2$ [4, Theorem 6.1]; Hong runs it to depth 10^8 without a stall.

2. Supports with rational sums

Theorem 1. *Let $b \geq 2$ be an integer and let $A \subseteq \mathbb{N}$ be infinite with $\sum_{a\in A} 1/a < \infty$. Then $\sum_{a\in A}(b^a-1)^{-1}$ is irrational.*

Erdős proved the case of pairwise coprime A and stated this extension without proof [3, p. 222]. The proof in [1, Theorem A.1] uses the displacements

$$\Delta_{b,A}(N) = \sum_{a\in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \qquad J_{b,A}(N) \in \mathbb{Z},$$

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which are positive because A is infinite. A rational value p/q would give $\Delta_{b,A}(N) \geq 1/q$ for every N. Averaging N over the multiples of $\operatorname{lcm}(1,\dots,t)$ and then letting $t \to \infty$ drives the average displacement to 0; reciprocal summability supplies the dominating sums for both limits. The Lean declaration is `irrational_erdosSupportSeries_of_summable_reciprocal` [5], and it has passed a Comparator replay against a statement written from Mathlib alone.

The introduction of [4] observes that if Erdős's unprinted statement holds, a rational value requires $\sum_{a\in A} 1/a = \infty$. Theorem 1 supplies that statement. A weighted version gives more. For a finite nonempty set P of primes, write $h(a) = \prod_{p\in P} p^{v_p(a)}$. If $\sum_{a\in A} h(a)/(a(2^{h(a)}-1)) < \infty$ for an infinite A, then X_A is irrational [1, Theorem 1.2], also with a Lean proof [6]. A support with a rational sum therefore satisfies

$$\sum_{a\in A} \frac{h(a)}{a\,(2^{h(a)}-1)} = \infty \qquad \text{for every finite nonempty set } P \text{ of primes.}$$

Since $h(a) = 1$ when a is coprime to every prime in P, this condition holds whenever the elements of A coprime to $\prod_{p\in P} p$ have divergent reciprocal sum. The candidate A_*, whose computed density is close to $1/2$ at every scale examined, is consistent with all of these conditions.

Proposition 2. *Let $d \geq 2$ and let $A \subseteq \{2,3,\dots\}$ be nonempty with $dA \subseteq A$ and $A \setminus dA$ finite. Then $\sum_{a\in A} 1/a < \infty$, so $X_A(b)$ is irrational for every integer $b \geq 2$.*

Proof. Put $B = A \setminus dA$. An element $a \in A \setminus B$ equals da' with $a' \in A$ and $a' < a$, so by descent every $a \in A$ equals $d^j b$ for some $b \in B$ and $j \geq 0$. Hence $\sum_{a\in A} 1/a \leq \frac{d}{d-1}\sum_{b\in B} 1/b < \infty$. The set A is infinite because $dA \subseteq A$, and Theorem 1 applies. □

For these supports Hong proves that $X_A(2)$ is transcendental [4, Proposition 9.2], a stronger conclusion at base 2. Proposition 2 adds irrationality at every other integer base.

3. The fair-digit measure

Let ν be the law of $\sum_{n\geq 2} \varepsilon_n w_n$ when $\varepsilon_2, \varepsilon_3, \dots$ are independent fair digits, as in [4, §8]. Let $E = \sum_{n\geq 1} w_n$ and let λ be Lebesgue measure.

Proposition 3. *$\nu = 2\lambda$ restricted to $C_2 \cap [0, E-1]$. In particular a subset of $C_2 \cap [0, E-1]$ is ν-null exactly when it is Lebesgue-null.*

Proof. For $n \geq 1$ let $K_n = \{\sum_{m\geq n} \varepsilon_m w_m : \varepsilon_m \in \{0,1\}\}$. Since $w_n > \sum_{m>n} w_m$, the set K_n is the disjoint union of K_{n+1} and $w_n + K_{n+1}$, so $\lambda(K_n) = 2\lambda(K_{n+1})$. As $\lambda(K_1) = 1$ [4, §2.1], [1, §C.2], this gives $\lambda(K_n) = 2^{1-n}$. In particular $K_2 = C_2 \cap [0, E-1]$, because $K_1 = K_2 \cup (1 + K_2)$ and $E - 1 < 1$. Fixing $\varepsilon_2, \dots, \varepsilon_{n-1}$ confines the sum to a translate of K_n, of Lebesgue measure 2^{1-n}; its ν-probability is $2^{2-n} = 2 \cdot 2^{1-n}$. Distinct choices give disjoint translates, and these translates generate the Borel sets of K_2, because $(\varepsilon_n)_{n\geq 2} \mapsto \sum_{n\geq 2} \varepsilon_n w_n$ is a homeomorphism from $\{0,1\}^{\{2,3,\dots\}}$ onto K_2. Hence $\nu = 2\lambda$ on K_2. □

The Lean development of the record proves the same identity with ε_1 included: the fair-digit law of $\sum_{n\geq 1} \varepsilon_n w_n$ is λ restricted to C_2 (theorem `fairCoding_pushforward_eq_volume_restrict`, built on 8 September 2026 with axioms `propext`, `Classical.choice` and `Quot.sound`). Proposition 3 is its restriction to $\varepsilon_1 = 0$.

Consequently [4, Theorem 8.5] holds for Lebesgue-almost every target in $C_2 \cap (0, E-1)$, which settles the point left open after that theorem. Rational targets form a Lebesgue-null set and stay outside its scope.

4. Uniform cap certificates

Hong calls a depth k *uniformly cap-certified* if for some $e \in [k+1, 2k-1]$ and some valid carry bound $C \geq 2$ the window $[k+1, e]$ is pinned for every choice of values $0 \leq S_q \leq d(q) - 2$ [4, §9.4]. By [4, Lemma 3.4] this happens exactly when the recursion

$$g_{e+1} = C, \qquad g_i = \Big\lfloor \frac{d(i) - 2 + g_{i+1}}{2} \Big\rfloor \quad (i \leq e) \tag{2}$$

reaches 0 at some $i \in [k+1, e]$. [4, Proposition 9.4] shows that a uniformly cap-certified depth $k \geq 7$ forces a prime $q \in (k, 2k)$ with $(q+1)/2$ prime, and notes that such pairs are not known to exist in every interval $(k, 2k)$. The hypothesis is never satisfied.

Proposition 4. *Let $C \geq 2$ and $e \geq 8$ be integers and define g by (2). Then $g_i \geq 1$ for every integer i with $8 \leq i \leq e$. Consequently no depth $k \geq 7$ is uniformly cap-certified.*

Proof. Suppose otherwise and let i be the largest integer with $8 \leq i \leq e$ and $g_i = 0$. Then $g_j \geq 1$ for $i < j \leq e+1$, since $g_{e+1} = C \geq 2$. Each step below uses $d(j) - 2 + g_{j+1} \leq 2g_j + 1$, which is (2) read as an inequality.

(i) From $g_i = 0$ and $g_{i+1} \geq 1$ we get $d(i) = 2$ and $g_{i+1} = 1$. So i is a prime with $i \geq 11$, and $i + 1 \leq e$ because $g_{e+1} \neq 1$.

(ii) From $g_{i+1} = 1$ and $g_{i+2} \geq 1$ we get $d(i+1) \leq 4$. The even number $i + 1 \geq 12$ is then $2q$ with q an odd prime: a multiple $4m$ with $m \geq 3$ has the five divisors $1, 2, 4, 2m, 4m$, and $2m$ with m odd and composite has $d(2m) = 2d(m) \geq 6$. Hence $d(i+1) = 4$, $g_{i+2} = 1$ and $i + 2 \leq e$.

(iii) The primes i and $q = (i+1)/2 \geq 7$ are prime to 3, so $3 \nmid i$ and $3 \nmid i + 1$, and therefore $3 \mid i + 2$. Write $i + 2 = 3m$ with $m \geq 5$ odd; its divisors include $1, 3, m, 3m$, so $d(i+2) \geq 4$. From $g_{i+2} = 1$ and $g_{i+3} \geq 1$ we get $d(i+2) = 4$, $g_{i+3} = 1$ and $i + 3 \leq e$.

(iv) Finally $i + 3 = 2(q+1)$ with q odd, so $4 \mid i + 3$, and $i + 3 \geq 16$. Its divisors include $1, 2, 4, (i+3)/2, i+3$, so $d(i+3) \geq 5$. But $g_{i+3} = 1$ and $g_{i+4} \geq 1$ give $d(i+3) \leq 4$, a contradiction. □

The obstruction sits in the three integers after the prime i, which are divisible by 2, 3 and 4. The caps $d(q) - 2 + (1 - b_q)$ that [4, §7] uses for a target with binary digits b_q are at least $d(q) - 2$, and (2) is monotone in the caps, so the conclusion holds for every target.

Corollary 5. *For every integer $m \geq 8$,*

$$U(m) = \sum_{t \geq 0} \big(d(m+t) - 2\big)\, 2^{-(t+1)} \geq 1.$$

Proof. Suppose $U(m) < 1$ and run (2) with $C = 2$ from a right end $e \geq m$. Repeated use of $\lfloor (n + \lfloor x \rfloor)/2 \rfloor = \lfloor (n + x)/2 \rfloor$, for integers n and reals x, gives

$$g_m = \Big\lfloor \sum_{t=0}^{e-m} \big(d(m+t) - 2\big) 2^{-(t+1)} + 2^{-(e-m)} \Big\rfloor \leq \lfloor U(m) + 2^{-(e-m)} \rfloor,$$

which is 0 once $2^{-(e-m)} < 1 - U(m)$. This contradicts Proposition 4. □

The carries of every representation of $1/2$ satisfy $c_m \leq \lfloor U(m) \rfloor$, by [4, Lemma 3.1] and $S_q \leq d(q) - 2$. A position with $U(m) < 1$ would therefore force $c_m = 0$ for every support. At each finite order r such reset positions exist: $U_r(m) < 0.61$ whenever $m \equiv 1 \pmod{L_r}$ [4, Lemma 4.1], and they drive the reset propagation of [4, Theorem 4.6]. Corollary 5 shows that the full series has none beyond 7. A direct computation of U for $m \leq 10^6$ finds exactly the positions $2, 3, 4, 5, 7$, and a computation of (2) with $C = 2$ for $i \leq 10^7$ finds zeros only at the same five positions. So the pointwise caps force no carry to vanish at any depth beyond 7, and a certificate for [4, Conjecture 6.3] needs more than the pointwise caps, for instance the relations among the S_q that come from a single support, as anticipated at the end of [4, §9.4].

5. Independent reproduction

The process to depth 10^8. We implemented the process of [4, §6] from its description, without the ancillary archive of [4]. The program computes the greedy prefix of length 64 in exact rational arithmetic, uses the carry bound $\overline{C}(k) = \lfloor (d(k) - 2 + \overline{C}(k+1))/2 \rfloor$ started from $3 \cdot 10^4$, takes at each depth the shortest pinned window $[k+1, k+l]$ for the bound $\overline{C}(k+l+1)$, and checks (1) at every depth. Its digits agree with the greedy expansion of $1/2$ computed in exact rational arithmetic through $n = 1500$. Table 1 compares the run with [4, §6.1 and Table 3].

Quantity, for $N = 10^8$	Hong [4]	This note
Depth reached without a stall	10^8	10^8
Density of A_* on $(N/2, N]$	0.50005	0.500045
Largest carry	196	196
Longest window, first depth	31, 43,177,919	31, 43,177,919
Longest run of consecutive elements	27	27 (from 43,177,920)
Depths with window ≥ 14, 18, 20, 22	195,868; 12,453; 3,140; 813	identical
Depths with window ≥ 24, 26, 28	198; 37; 7	identical

Table 1. The process for $1/2$ to depth 10^8.

The run also gives $|A_* \cap [1, 10^8]| = 50{,}009{,}676$, a longest gap of 24 (the integers 83,764,427 to 83,764,450 lie outside A_*), and windows of length at least 30 at exactly two depths. It took about two minutes on an Apple M4 machine shared with other jobs.

An earlier program in the Plectis record, written independently of both [4] and the present code, computes the same support by doubling the depth in blocks with 256-bit tail windows [2]. Both programs give $|A_* \cap [1, 13{,}548{,}057]| = 6{,}782{,}259$ and the same largest carry, 123, on that range.

Pinning lengths. For the finite orders of [4, Table 2] we computed ℓ_r directly from [4, Lemma 3.4]: for every right end e modulo L_r the recursion with $g_{e+1} = C_r$ and caps d_r runs leftwards to its first zero, and ℓ_r is the longest window found. The caps come from a table of d_{22} with period L_{22} together with divisibility tests for $23 \leq j \leq r$. The program reproduces ℓ_r for every $2 \leq r \leq 24$. For $r = 23$ and $r = 24$ the maximum 1023 is attained at six right ends, the first being 871,374,004, as reported in [4, §5.1]. These two rows, which [4] records as resting on a single complete scan, now rest on two.

Code and provenance

The programs, their outputs and the commands used are at https://github.com/wcook04/plectis-erdos/tree/3cc8df2faad681b26f6694772167d13ef26dac49/research/experiments/erdos257.

Authorship and AI use. Will Cook directed this work. AI agents (Claude, Anthropic) wrote the programs, ran the computations and drafted the text; Cook did not independently verify every claim.

References

[1] W. Cook, *Weighted support criteria for reciprocal Mersenne subseries*, Plectis research note for Erdős Problem 257, https://github.com/wcook04/plectis-erdos/blob/a042843b67cefd74ce71a2fd9e995a88f7897172/paper/257/erdos-257-mersenne-support-subseries.pdf.

[2] W. Cook, *Reciprocal Mersenne subseries*, Plectis research record for Erdős Problem 257, and its computations, https://github.com/wcook04/plectis-erdos/tree/3cc8df2faad681b26f6694772167d13ef26dac49/research/experiments/erdos257.

[3] P. Erdős, On the irrationality of certain series, *Math. Student* **36** (1968), 222–226.

[4] C. M. Hong, *On the rationality of reciprocal Mersenne subsums: carry pinning, quasi-polynomial certificates, and exact computation*, Zenodo, 2026, https://doi.org/10.5281/zenodo.23287959.

[5] Lean declaration `Erdos249257.irrational_erdosSupportSeries_of_summable_reciprocal`, https://github.com/wcook04/plectis-erdos/blob/436f55ebdafa67e4af0fff79f621c13f2ded12bf/lean/Erdos249257/AllBaseReciprocalSupportIrrationality.lean#L395.

[6] Lean declaration `ErdosProblems.Erdos257.PaperCompleteR8.divisibilityWeightedClaim`, https://github.com/wcook04/plectis-erdos/blob/436f55ebdafa67e4af0fff79f621c13f2ded12bf/lean/ErdosProblems/Erdos257/PaperCompleteR8/WeightedReturn.lean#L120.